# A Computationally Feasible Framework for Causal Probabilistic Explanation

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## Abstract

Causal attribution and explanation are mainstays of philosophical, scientific, and policy analysis. Some current frameworks, however, such as actual causality, suffer from an inability to scale causal reasoning to large problems, while attribution methods that do scale—such as SHAP or LIME—are not causally aware. To address this two-fold problem, we present Probabilistic Causal Impact (PCI). Inspired by Halpern and Pearl's actual causality and Pearl's probability of necessary and sufficient causation, PCI provides tractable causal explanations in causal probabilistic machine learning models. By the same token, PCI provides explanations for events in the world, insofar as the model adequately represents real mechanisms. By re-conceptualizing explanation as an estimation problem on an expanded, causal probabilistic model, PCI allows for the use of well-established approximation algorithms, unlocking computationally tractable, truly causal explanations for large models. We test PCI on scaled-up, but canonical test cases for actual causation involving overdetermination and undercutting, finding that we recover results deemed intuitive by the literature, and illustrate that the correct behavior is preserved in the continuous version of the problem. We apply PCI to a real-life causal spatio-temporal model with complex machine-learned components trained on millions of data points. In this setting, we compare PCI to the non-causal (yet scalable) attribution method SHAP, and find that our approach yields explanations that agree with domain expert opinion, while SHAP does not.


**Keywords:** Probabilistic Causal Models, Explanation, Responsibility, Actual Causality, Probability of Necessity and Sufficiency

## 1. Introduction

Causal attribution is a central reasoning task in many important problems, such as gauging the efficacy of medical treatments (Shepherd et al., 2024), pinpointing drivers of economic growth (Cheshire and Magrini, 2009), root cause analysis in automated systems (Papageorgiou et al., 2022), or assessing the impact of public policy (Heckman and Pinto, 2022). In general, we may want to understand why an adverse outcome occurred to prevent it from happening again, or understand how and why a predicted, counterfactual positive outcome can be brought about. However, in the absence of scalable, formal causal attribution tools, analysts often resort to either ad-hoc methods, such as mechanistic reasoning and finding strong correlation, historical comparison, domain expertise, or non-causal attribution like Granger causality (Granger, 1969) and SHAP (Lundberg and Lee, 2017). Even researchers with an advanced command of statistical and experimental methods may not possess a thorough understanding of how to ascribe cause (Shepherd et al., 2024), and would thus benefit from scalable automation. Here we present a tool for accurate and tractable causal attribution that can be applied to a wide array of causal, probabilistic, machine-learned models.

Current methods for causal attribution that focus on token-level explanations of particular events, such as actual causality (Halpern, 2016), involve intractable combinatorics without clear approximating machinery (Eiter and Lukasiewicz, 2002). Explanation methods that *do* scale are either non-causal (Lundberg and Lee, 2017) or amount to local, gradient-based sensitivity analyses (Butler et al., 2022). For example, Lundberg and Lee's SHAP method may assign high attribution to a feature simply because it correlates with a true cause, not because altering said feature would change the outcome. Butler et al. in their Differential Causal Effect (DCE) rely on gradients for attribution, which requires model differentiability, and does not capture effects that require more than infinitesimal changes in potential causes (admittedly, the authors are only after rates of change of treatment, which is conceptually different from responsibility attribution).

In contrast, we introduce probabilistic causal impacts (PCI), which exploit probabilistic understandings of context and causation to frame attribution as a turnkey estimation problem. In particular, we generalize Pearl's formulation of the "probability of necessary and sufficient causation" (Pearl, 2022) for arbitrary variable domains, and additionally incorporate causal context in the form of "witnesses" as understood by actual causality researchers (Halpern, 2016). Inspired by the study of "descriptive normality" (Halpern and Hitchcock, 2015; Icard et al., 2017) — how statistical probabilities of counterfactuals influence causal attributions — we characterize attribution as the expectation of a customizable causal impact function taken with respect to user-defined distributions that assign probabilities to counterfactual worlds. PCI combines a variety of intuitions from the literature, including taking expectations over exogenous inputs (cf. Halpern (2016) on "blame", def. 6.2.1; and implicitly by Pearl's (2022) probabilities of causation), weighting by the probability of alternative values for the supposed cause (Beckers and Vennekens, 2015, def. 8), aggregating the probabilities of all valid witness worlds (Beckers and Vennekens, 2015, §6.2), or weighting by the complexity of the least complex witness set (Halpern, 2016, on "responsibility", def. 6.1.6). In Appendix **??**, we discuss related concepts in more detail. Unfortunately, taken individually, none of these existing approaches achieves the scalability and generality required for modern causal attribution needs, thereby necessitating the development of PCI.

The mathematical and computational tooling we develop in this paper is designed for a wide range of causal probabilistic models; we place no strong requirements on the form of the model, distributions, structural equations, or types of variables that can be used. Indeed, such models can comprise standard machine learning components like neural networks or Gaussian processes. With this breadth of application, and the turnkey nature of our measure of causal attribution, PCI hopes to provide *truly causal* attribution and explanation tooling for a wide array of users and applications.

To validate and better understand the approach, and to illustrate how our method scales to far larger models than does exact actual causality computation, we take the following steps:

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1. We draw theoretical connections to existing (but intractable) attribution strategy, actual causality in Section **??**, where we prove that, under certain constraints, the diagnoses offered by actual causality and by PCI coincide.
2. In Section **??**, we test PCI on scaled-up, but canonical failure cases for actual causation involving overdetermination and undercutting, finding that we recover results deemed intuitive by the literature,
3. In Section **??** we point also to the correct behavior in the continuous version of the problem.
4. We also delegate a more extensive conceptual discussion of the relation between PCI , actual causality, and probabilities of necessity, sufficiency and necessity and sufficiency to Appendix **??**.

5. In Section **??** we describe results on a causal spatio-temporal model with complex machine-learned components trained on millions of data points. In this setting, we compare to the non-causal but scalable attribution method SHAP, and find that our approach yields explanations which agree with domain expert opinion, while SHAP does not.
6. Additionally, in Appendix **??**, we compare the local, sensitivity-analysis-style attribution from Butler et al. (2022), demonstrating cases where PCI better reveals causality.

## 2. Causal Impact: motivations

The central difficulty in causal attribution is not computing counterfactuals — it is knowing which counterfactuals to compute, and recognising when a cause is blocked by context rather than absent altogether. The closest predecessors of this approach, taken individually, fail to capture this: actual causality handles context sensitivity but is computationally intractable and not general enough; the probability of necessity and sufficiency scales well, but lacks context sensitivity. This section builds intuition for both limitations through a running example, setting up the design choices that motivate PCI in Section 3.

### 2.1. Actual Causality

Consider the following (deterministic, discrete) model:

**Example 1 (Bob at Old Boys' Club Bank)** *Bob applied for a home mortgage loan with the local branch of Old Boys' Club Bank (OBCB), and his application was denied. The OBCB policy requires applicants to pass a credit score check, but it only performs this check if the applicant is male; all non-male applicants are rejected outright. Thus, the procedure is:*

$$\begin{aligned} \mathit{check} &= (\mathit{gender} = \mathit{male}) \\ \mathit{check\text{-}failed} &= \mathit{check} \wedge (\mathit{credit} = \mathit{bad}) \\ \mathit{loan} &= \neg \mathit{check\text{-}failed} \wedge (\mathit{gender} = \mathit{male}) \end{aligned}$$

Why was Bob's loan application denied? An intuitive explanation is that his credit was bad—had it been good, the outcome would have been different. In contrast, his gender does not appear to be responsible for the rejection: had it been different, his application still would have been denied (albeit this time due to $\mathit{check} = F$).

This suggests a counterfactual approach to feature responsibility attribution: perhaps we can identify the features responsible for a particular outcome by asking counterfactual questions—would the outcome have been different if a specific feature had a different value? This is called the *but-for clause*: Bob wouldn't have been denied the loan but for his bad credit.

Concretely, consider a model $M$ with *exogenous variables* $\mathbf{U}$, which are the source of stochasticity, and *endogenous variables* $\mathbf{V}$, which are influenced by other variables in the model. Let $\mathbf{S} \subseteq \mathbf{V}$ be the variables we suspect to be causally responsible (call them *causal suspects*), and $Y \in \mathbf{V}$ be an outcome of interest. Suppose we consider only one suspect $S \in \mathbf{S}$ and mark factual values with $^\star$. Under a setting $\mathbf{U} = \mathbf{u}$ of the exogenous variables, $M$ produces $S = s^\star$ and $Y = y^\star$, denoted $(M, \mathbf{u}) \models S = s^\star, Y = y^\star$. We call this the *factual* scenario. An intervened model where a variable $S$ is set to an alternative value $s'$ is denoted $(M, \mathbf{u})[S \leftarrow s']$.

A first attempt at defining a counterfactual notion of causal responsibility is:

**Definition 1 (But-for, single)** *$S = s^\star$ is causally responsible for $Y = y^\star$ in $(M, \mathbf{u})$ if and only if $(M, \mathbf{u}) \models S = s^\star, Y = y^\star$ and $(M, \mathbf{u})[S \leftarrow s'] \not\models Y = y^\star$ for some $s' \neq s^\star$.*

In a given context, $S$ is causally responsible for an outcome only when intervening on $S$ with some alternative value would change that outcome. Bob's credit score satisfies this condition: had his credit score been higher, the loan might have been approved. However, this cause test is too coarse.

Consider another case:

**Example 2 (Alice at OBCB)** *Alice is a woman with bad credit. She applies for a loan with OBCB and is rejected.*

Why was Alice denied the loan? Under the but-for clause for single cause candidates, neither her gender nor her credit score are responsible. No matter how high her credit score, her application would still have been denied because the bank does not loan to women. Had her gender been different, she would still have been denied for having bad credit.

To address this example, we could consider sets of factors as causes:

**Attempt 1 (But-for, sets)** *Variables $\mathbf{C} \subseteq \mathbf{S}$ with factual values $\mathbf{C} = \mathbf{c}^\star$ are causally responsible for $Y = y^\star$ in $(M, \mathbf{u})$ if and only if $(M, \mathbf{u}) \models \mathbf{C} = \mathbf{c}^\star, Y = y^\star$ and $(M, \mathbf{u})[\mathbf{C} \leftarrow \mathbf{c}'] \not\models Y = y^\star$ for some $\mathbf{c}'$ everywhere different from $\mathbf{c}^\star$, where no proper subset of $\mathbf{C}$ has this property.*

The minimality condition reflects our preference for explanations that include only relevant properties.

Under this generalization, the set $\mathbf{c}^\star = \{\mathit{gender} = \text{female}, \mathit{credit} = \text{bad}\}$ is jointly responsible for Alice's application being denied because there exists $\mathbf{c}' = \{\mathit{gender} = \text{male}, \mathit{credit} = \text{good}\}$ that would change the decision and $\mathbf{c}^\star$ is minimal. But what can we say about the individual features? We could say that the whole set bears responsibility but individual features do not, or that any feature in a cause set is partially and equally accountable. Both approaches have issues, as they imply the features are equally responsible. This conflicts with our intuition: OBCB did not evaluate Alice's credit score, so her denial was caused by her gender.[1]

Prior work suggests a way out of this difficulty: hold some of the context in which counterfactuals are evaluated fixed, calling such fixed variables *witnesses*. Let us focus on the mediator $\mathit{check\text{-}failed}$ between $\mathit{gender}$ and $\mathit{loan}$. If we fix this variable at the factual value by intervening, then counterfactually, Alice would have been granted the loan if she were male. However, there is no context fixed at the actual value such that, had her credit score been higher, she would have been granted the loan. This motivates the following formalization:

**Definition 2 (Actual cause)** *Let $\mathbf{C} \subseteq \mathbf{S}$ be a cause set, $\mathbf{W} \subseteq \mathbf{V}$ be a set of possible witnesses. $\mathbf{C} = \mathbf{c}^\star$ is causally responsible for $Y = y^\star$ in $(M, \mathbf{u})$ just in case there is a $\mathbf{T}$ (an actual witness set to be held fixed) s.t. $\mathbf{T} \subseteq \mathbf{W}, \mathbf{T} \cap \mathbf{C} = \emptyset$ and $(M, \mathbf{u}) \models \mathbf{C} = \mathbf{c}^\star, Y = y^\star$ and $(M, \mathbf{u})[\mathbf{C} \leftarrow \mathbf{c}', \mathbf{T} \leftarrow \mathbf{t}^\star] \not\models Y = y^\star$, for some $\mathbf{c}'$ distinct everywhere from $\mathbf{c}^\star$, where no proper subset of $\mathbf{C}$ satisfies this property.*

1. Examples of this sort, involving both over-determination and undercutting of causal impact, abound. Suppose Jim is both stabbed and poisoned and dies as a result. Because stabbing works quickly, it kills Jim before the poison takes effect. In this case, it's the stabbing that caused his death. Yet analogous but-for clauses fail to break the symmetry here as well.

This definition extends but-for causality with a witness set $\mathbf{T}$ that is fixed to its factual value by an intervention. Under this definition, to determine the cause of an outcome, we need to find a cause set $\mathbf{C}$, alternative values $\mathbf{c}'$ (trivial if the variables are binary and less so if they are categorical or continuous), and a witness set $\mathbf{T}$. $\mathbf{C}$ and $\mathbf{T}$ are drawn from sets of size at most $2^{|\mathbf{V}|}$, so a brute-force computation of these sets is expensive.

Putting computational difficulties aside, we have so far considered deterministic models, but would like to capture probabilistic models as well, so we now consider a stochastic variant of the OBCB problem.

### 2.2. Probability of Necessity and Sufficiency

**Example 3 (OBCB, stochastic)** *The bank performs credit checks for a randomly selected 20% of women and skips checks for a randomly selected 10% of men. Unchecked applicants are rejected. Furthermore, if a man's credit check fails, he is nevertheless accepted 5% of the time and if a woman's credit check succeeds, she is denied 10% of the time. We assume that the population is evenly divided into male and female and good and bad credit. Formally, the model, using 1 for male and 1 for good credit, is:*

$$
\textit{loan-prob} = \begin{array}{c|cc}
 & \textit{check-failed}{=}0 & \textit{check-failed}{=}1 \\ \hline
\textit{gender}{=}0 & 0.9 & 0 \\
\textit{gender}{=}1 & 1 & 0.05
\end{array}
$$

$$
\begin{aligned}
\textit{credit} &\sim \mathrm{Bern}(0.5) \\
\textit{gender} &\sim \mathrm{Bern}(0.5) \\
\textit{check} &\sim \mathrm{Bern}((1-\textit{gender})\cdot 0.2 + \textit{gender}\cdot 0.9) \\
\textit{check-failed} &= \textit{check}\cdot(1-\textit{credit}) \\
\textit{loan-if-checked} &\sim \mathrm{Bern}(\textit{loan-prob}[\textit{gender}, \textit{check-failed}]) \\
\textit{loan} &= \textit{loan-if-checked}\cdot\textit{check}
\end{aligned}
$$

The witness mechanism works well for deterministic models, but actual causality in its standard form offers no straightforward way to handle probabilistic ones. Before extending the witness idea to the stochastic setting, it is instructive to examine a probabilistic approach that scales more naturally: Pearl's probabilities of necessity and sufficiency (Pearl, 2022). This will let us identify what probabilistic machinery is needed, and what context-sensitivity is lost when witnesses are absent — motivating the synthesis that PCI provides. The following definitions allow for probabilistic models, but assume that all variables are Boolean (PCI will relax this assumption, allowing for continuous variables). Pearl considers two ways that a cause may relate to an effect: it may be necessary or sufficient for the effect. A cause is necessary if, counterfactually absent the cause, the effect is unlikely to occur; it is sufficient if the effect is counterfactually likely in the presence of the cause. We now formalize these ideas and discuss them in the context of the running example.

**Definition 3 (Probability of Necessity)** *Let the probability of necessity be the probability that the outcome $Y = y$ would not have occurred in the absence of $C = c^\star$, given that both occurred:*

$$PN(C = c^\star, Y = y^\star) = P(Y_{c'} \neq y^\star \mid C = c^\star, Y = y^\star)\,,$$

*where $Y_{c'}$ is the value of $Y$ after intervening on $C$ with $c' \neq c^\star$.*

The probability of necessity is less specific to a particular situation than the actual cause test discussed above. For example, we can consider the probability that being male causes someone to fail to get a loan, but we can't consider the probability that being male causes Bob specifically (who we know has bad credit) to fail to get a loan.

This is messy, cleanup

To compute

$$PN(gender = \text{male}, loan = F) = P(loan_{gender=\text{female}} = T \mid gender = \text{male}, loan = F),$$

we first determine that $P(credit^\star = \text{good}) = \frac{20}{211} \approx 0.09$ using the condition (men with good credit are rarely rejected). The probability of getting a loan in the counterfactual case is then $P(credit^\star = \text{good}) \times 0.2 \times (1 - 0.1) = \frac{18}{1055} \approx 0.02$. We find that $PN(gender = \text{female}, loan = F) \approx 0.43$ and $PN(credit = \text{bad}, loan = F) \approx 0.53$; bad credit has a notably higher necessity score than being female. Both exceed $PN(gender = \text{male}) \approx 0.02$, reflecting that female gender and bad credit are each more likely to necessitate rejection. The value for $gender = \text{female}$ remains below $0.5$ because many rejected females also have bad credit, which would prevent loan approval even in the counterfactual world where gender is male. However, we know that having low credit and being female are actually quite likely to cause a loan rejection on their own. For Alice specifically, $PN(gender = \text{female}, loan = F) = P(loan_{gender=\text{male}} = T \mid gender = \text{female}, credit = \text{bad}) = 0.045$: if she were male, a credit check would be performed (90% probability), and despite bad credit she would have a 5% chance of approval. For her credit, the counterfactual is $P(loan_{credit=\text{good}} = T \mid gender = \text{female}, credit = \text{bad}) = 0.2 \times 0.9 = 0.18$. Bob's credit necessity score is 0.90: had his credit been good, a check would have been performed and he would almost certainly have been approved.

Notice how Bob's PN score for credit gets high necessity (0.9), while Alice's is lower (0.18). This partially captures the intuition that Bob's bad credit is more responsible for his rejection than Alice's. But while Bob's PN score for gender is 0, which perhaps partially captures the intuition that his gender is not responsible for his rejection, Alice's PN score for gender is only 0.045, which is not that different, and does not capture the intuition that her gender did play a significant role in her rejection. To see the full picture, we must consider another quantity: the probability of sufficiency.

**Definition 4 (Probability of Sufficiency)** *Let the probability of sufficiency be the probability that the outcome $Y = y^\star$ would have occurred in the presence of $C = c^\star$, given that $C = c^\star$ did not occur:*

$$PS(C = c^\star, Y = y^\star) = P(Y_{c^\star} = y^\star \mid C = c',\ Y = y')$$

*where $Y_{c^\star}$ is the value of $Y$ after intervening on $C$ with $c^\star \neq c'$.*

To query whether the factual cause was sufficient, we condition on the counterfactual: we assume the cause did *not* take its factual value and the outcome did *not* occur, then intervene to restore the factual cause value and ask whether the outcome now occurs. Intuitively, this asks: if the cause had been absent and things had gone differently, would reinstating the cause have brought the outcome about? Let's look at the resulting values.

Computing the probabilities of sufficiency, we find that $PS(gender = \text{male}, loan = F) \approx 0.1$, $PS(gender = \text{female}, loan = F) \approx 0.83$, and $PS(credit = \text{bad}, loan = F) \approx 0.96$. These values tell us that being female or having bad credit are both on their own sufficient for a loan rejection, with bad credit being slightly more sufficient. For Alice, both variables achieve $PS = 1$: being female is fully sufficient to cause denial as the model offers no path to approval for women with bad credit, and having bad credit forecloses approval entirely, since $P(loan = 1 \mid gender = \text{female}, credit = \text{bad}) = 0$. Bob's gender yields an undefined PS: computing it requires conditioning on Bob being female, having bad credit, and succeeding, a conjunction the model assigns probability zero, making the conditional undefined. Bob's credit yields $PS = 0.955$: restoring bad credit leads to denial in $95.5\%$ of worlds, not certainty, because even with bad credit there remains a $4.5\%$ chance of approval ($90\%$ probability of being checked, times the $5\%$ exception rate for males who fail the check). These cases illustrate that PS can fall short of $1$ even when a factor is strongly sufficient, and can be undefined when the conditioning event is near-impossible.

To somehow combine insights provided by the two scores, we would like to find causes that are both necessary and sufficient. This notion is formalized as follows:

**Definition 5 (Probability of necessity and sufficiency)** *Let the probability of necessity and sufficiency be the probability that the outcome $Y = y^\star$ would have occurred in the presence of $C = c^\star$ and that it would not have occurred under the intervention $C = c'$:*

$$PNS(C = c^\star, Y = y^\star) = P(Y_{c^\star} = y^\star, Y_{c'} \neq y^\star)$$

When we compute these probabilities for our example, we find that $PNS(gender = \text{male}, loan = F) \approx 0.01$, $PNS(gender = \text{female}, loan = F) \approx 0.39$, and $PNS(credit = \text{bad}, loan = F) \approx 0.52$. While these probabilities reflect the relative causal impact of $gender$ and $credit$, they do not provide entirely satisfying answers. Table 1 reveals the problem directly: for Alice, $PNS(gender) = 0.045 < PNS(credit) = 0.18$, so PNS ranks credit as more causally responsible than gender. Yet the bank never evaluated Alice's credit—the $check$ variable was $F$ throughout. PNS has no mechanism to register this: it asks counterfactually what would happen if credit were different, without fixing the mediating variable $check\text{-}failed$ that actually blocked credit's causal path.

As an attempt to help Alice better understand her particular situation, we might consider a modification of PNS with additional conditioning:

$$PNS_c(C = c^\star, Y = y^\star \mid F = f^\star) = P(Y_{c^\star} = y^\star, Y_{c'} \neq y^\star \mid F = f^\star)$$

Using this definition, we can compute $PNS_c(gender = \text{female}, loan = F \mid credit = \text{bad}) = 0.045$ and $PNS_c(credit = \text{bad}, loan = F \mid gender = \text{female}) = 0.18$. Conditioning does not solve the problem described above—Alice's credit was in reality never evaluated—because the interventions override the mediating variable that tracks whether her credit has been checked. When we compute $PNS_c(credit = \text{bad}, loan = F \mid gender = \text{female})$, conditioning on gender being female is consistent with the factual—but when we then intervene to set credit to good, this intervention propagates downstream and changes $check\text{-}failed$, overriding what we learned from conditioning on Alice's factual outcome. The context we fixed is undone by the very intervention meant to probe it. We will not consider this use of conditioning further, and this observation motivates the need for intervention-based (as opposed to conditioning-based) context-sensitivity, which is what the witness mechanism provides.

Table 1: Individual-level $PN$, $PS$, and $PNS$ for Alice and Bob in Example 3, with $loan = F$ as the outcome. Values are conditioned on each subject's factual context (female, bad credit for Alice; male, bad credit for Bob). A dash marks an entry that is undefined because the conditioning premises are inconsistent with the model.

| | Alice | | | Bob | | |
|---|---|---|---|---|---|---|
| | $PN$ | $PS$ | $PNS$ | $PN$ | $PS$ | $PNS$ |
| $gender$ | 0.045 | 1.00 | 0.045 | 0 | — | 0 |
| $credit$ | 0.18 | 1.00 | 0.18 | 0.90 | 0.955 | 0.86 |

### 2.3. Path forward

What should we conclude from this investigation of actual causality and the probability of necessity and sufficiency? First, we need a way to treat probabilistic models, and the probability of necessity and sufficiency gives us a tool for doing that. However, PNS is not as specific as we would like because it does not capture relevant features of individual units that are not part of the cause. Actual causality offers more specificity, but requires modification to deal with probability and continuous variables, and is computationally expensive to compute.

PCI addresses both limitations by combining the two approaches. To see the intuition, consider Alice's case once more. When PCI evaluates gender's causal impact on her denial, it considers counterfactual worlds in which her gender is changed, while drawing witness sets from a distribution that includes $check\text{-}failed$ at its factual value of $F$ — the credit check was never performed. In those worlds, changing gender to male, with $check\text{-}failed$ held fixed at $F$, leads to loan approval: a large counterfactual shift, and so high attribution. When PCI evaluates credit's causal impact, it searches for witness sets that, held fixed, would make a change to good credit matter for Alice — but no such context exists, because the credit check was never run. The distribution over witness sets therefore assigns low weight to any world in which credit is relevant for Alice, and her credit receives low attribution. Unlike actual causality, PCI does not ask whether *some* witness set validates a counterfactual dependency; it integrates over the distribution of all witness sets, weighting each by its probability. Unlike PNS, it inherits the context-sensitivity of actual causality through this integration. The formal definitions in Section 3 make this expectation precise — but the intuition is the one we have just seen: PCI inherits the witness mechanism of actual causality and the probabilistic language of PN/PNS, and combines them into a tractable estimation problem.

## 3. Causal Impact: definitions

Explanation and causal attribution seek to connect a potentially causal *event* $X = x$ to an outcome event $Y = y$.[2] In this section, we present a formal development, but the reader is invited to familiarize themselves with Appendix **??** for a pedagogical explanation that connects our approach to Pearl's probabilities of causation and Halpern's actual causality.

We define causes in the context of probabilistic structural causal models (Pearl, 2009).

2. This is quite different from type-level causal queries, where one is interested in some group-level attribution, as when one estimates average treatment effect or conditional average treatment effect.

**Definition 6 (Probabilistic Structural Causal Model)** *A probabilistic structural causal model is a tuple*

$$M = \langle \mathbf{U}, \mathbf{V}, \mathbf{F}, P_{\mathbf{U}} \rangle,$$

*where* $\mathbf{V} = \{V_1, \ldots, V_n\}$ *is a finite set of endogenous variables,* $\mathbf{U} = \{U_1, \ldots, U_n\}$ *is a set of exogenous variables, and* $\mathbf{F} = \{f_1, \ldots, f_n\}$ *is a set of structural assignments*

$$V_i := f_i(\mathbf{Pa}_i, U_i), \qquad i = 1, \ldots, n,$$

*with* $\mathbf{Pa}_i \subseteq \mathbf{V} \setminus \{V_i\}$ *denoting the parents of* $V_i$*. The distribution* $P_{\mathbf{U}}$ *is a probability measure on* $\mathrm{dom}\,(\mathbf{U})$*, and the assignments in* $\mathbf{F}$ *are assumed to be acyclic and to have a unique solution for* $\mathbf{V}$ *given* $\mathbf{U}$.

**Definition 7 (Interventions)** *Given a model* $M = \langle \mathbf{U}, \mathbf{V}, \mathbf{F}, P_{\mathbf{U}} \rangle$*, an intervention on variables* $\mathbf{X} = \{X_1, \ldots, X_k\} \subseteq \mathbf{V}$ *with values* $\mathbf{x} = (x_1, \ldots, x_k)$ *replaces the structural assignments for* $X_i$ *by* $X_i := x_i$*, where* $i = 1, \ldots, k$*. We denote this intervention by either* $[\mathbf{X} \leftarrow \mathbf{x}]$ *or* $do(\mathbf{X} = \mathbf{x})$.

Because PCI supports arbitrary variable types, it will be helpful to define a measure theoretic object corresponding to interventional outcomes. Because outcomes are a deterministic function of exogenous noise, this will be a Dirac measure.

**Definition 8 (Interventional Law)** *Let* $M = \langle \mathbf{U}, \mathbf{V}, \mathbf{F}, P_{\mathbf{U}} \rangle$ *be a probabilistic structural causal model and let* $Y \in \mathbf{V}$*. For an intervention* $\mathrm{do}(\mathbf{X} = \mathbf{x})$*, let* $Y' : \mathrm{dom}\,(\mathbf{U}) \to \mathrm{dom}\,(Y)$ *denote the measurable map induced by the modified structural assignments. The interventional law of* $Y$ *given fixed* $\mathbf{u} \in \mathrm{dom}\,(\mathbf{U})$ *is the probability measure on* $\mathrm{dom}\,(Y)$ *defined by*

$$P(Y \in A \mid \mathbf{u}, \mathrm{do}(\mathbf{X} = \mathbf{x})) \;:=\; \mathbb{I}\big\{Y'(\mathbf{u}) \in A\big\}, \qquad A \subseteq \mathrm{dom}\,(Y).$$

We define a *causal set* to be a set of assignments to the variables of a probabilistic structural causal model. In what follows, causal sets will take on multiple roles. At times, they will represent sets of *potential* causes—whose subsets may be intervened upon—in which case, we will call them suspects. We denote the set of all such potential causes by $\mathbf{S}$. When referring to specific sets we *in fact* intervene on, we use $\mathbf{C}$, calling them *active suspects*.

**Definition 9 (Causal set)** *Given a model* $\langle \mathbf{U}, \mathbf{V}, \mathbf{F}, P_{\mathbf{U}} \rangle$*, a causal set is a set of pairs* $X = x$ *where* $X \in \mathbf{V}$ *and* $x \in dom(X)$.

Analogously, some variables will serve as *potential* elements of the context to be kept fixed at the actual value, denoted by $\mathbf{W}$, and called *witnesses*; when referring to context variables that *in fact* are held fixed in a particular case, we use $\mathbf{T}$ and call them *active witnesses*.

To evaluate the causal impact of a variable on an outcome, we consider two interventional regimes: the *necessity regime*, in which we intervene on the variables $\mathbf{C} \subseteq \mathbf{S}$ to have an alternative value, and the sufficiency regime, in which we intervene so that $\mathbf{C}$ takes on factual values. Moreover, we hold some elements $\mathbf{T} \subseteq \mathbf{W}$ of the context fixed (i.e., intervened on to retain factual values, despite intervention on $\mathbf{C}$). When looking for an explanation, we perform a search across possible cause sets and possible contexts. In each case, attribution increases when the outcome is far from the factual value in the necessity world and decreases when it is far from the factual value in the sufficiency world.

To answer a causal responsibility query, we marginalize over two families of auxiliary variables: one pertains to causally responsible sets of variables and another determines context sets to hold fixed. The set of all candidate cause nodes will be called *suspects* and denoted by $\mathbf{S}$. Subsets of $\mathbf{S}$, when considered intervened in a particular search, will be called *active suspects* and denoted by $\mathbf{C}$. The candidate context variables are called *witnesses* and denoted by $\mathbf{W}$. In a particular search step, some subset of $\mathbf{W}$ will be, in fact, intervened to be kept at the factual values; we call those *active witnesses* and denote them by $\mathbf{T}$. Overlap is handled by preferring suspect activation (i.e., $\mathbf{T}$ is pruned of variables that belong to $\mathbf{C}$.

We expand an original model of interest with auxiliary sites that correspond to the moves involved in the search.

**Definition 10 (Variable Selection Distribution)** *Let $\mathbf{X}$ be a finite set of random variables and $2^{\mathbf{X}}$ be its power set. Let $\mathcal{Y}$ be a random set of random variables that is distributed according to $\Gamma(2^{\mathbf{X}})$ such that $\mathbf{X} \supseteq \mathbf{Y} \sim \Gamma(2^{\mathbf{X}})$ and probability mass function $p^{\Gamma}(\mathbf{Y}) = \mathbb{P}(\mathcal{Y} = \mathbf{Y})$.*

In particular, we will be using variable selection distributions for the powerset of causal suspects $\mathbf{S}$ and for the powerset of witnesses $\mathbf{W}$. One simple pattern of suspect and witness selection can be derived by setting size limits on the number of activated suspects or witnesses, and sampling uniformly otherwise.

**Example 4 (Cardinality-Constrained Uniform Selection)** *Let $\mathbf{X}$ be a candidate set of random variables and let $k \sim \mathrm{Unif}\{J, \dots K\}$ with $1 \leq J < K \leq |\mathbf{X}|$. A variable selection $\mathbf{Y} \sim \Gamma(2^{\mathbf{X}})$ is sampled from a cardinality-constrained uniform selection distribution $\Gamma(2^{\mathbf{X}})$ with probability mass function*

$$p^{\Gamma}(\mathbf{Y}) \propto \mathbb{I}[|\mathbf{Y}| = k].$$

Causal attribution hinges on questions such as "if $X$ had, contrary to fact, been fixed to $x' \neq x$ instead of $x$, what would the outcome have been?" For non-binary causes, this is ambiguous. To resolve this in a way that does not require committing to a single, contrastive causal event (Kawakami et al., 2024), we rely on a *distribution* over alternative values.

We will assume the model provides us with a family of interventional distributions/densities (SCMs are an example of a model class satisfying this requirement), and explain how those are to be used to build up expanded distributions needed for causal impact estimation.

**Definition 11 (Alternative Value Distribution)** *Let $\mathbf{X}$ be a set of random variables with domain $\mathrm{dom}\,(\mathbf{X})$ and let $\mathbf{X} = \mathbf{x} \in \mathrm{dom}\,(\mathbf{X})$ be a factual event. An* alternative value distribution *$\Delta(\mathbf{x})$ is a probability measure on $\mathrm{dom}\,(\mathbf{X})$ such that $\mathbf{x} \notin \mathrm{supp}(\Delta(\mathbf{x}))$.*

One natural alternative value distribution results from taking the one currently embodied in the model and excising it around the factual setting. Then, the semantics for "alternative value" means "a sufficiently different value, following the original distribution otherwise". In the discrete case, of course, the analogue is simply the probability mass function conditioned on the factual value not holding.

**Example 5 ($\epsilon$-Excised Distribution)** *Assume $\mathbf{X}$ is $\mathbb{R}^d$-valued with density $p$. Given a factual event $\mathbf{X} = \mathbf{x}$, define the $\epsilon$-excised alternative value distribution $\Delta_{\epsilon}(\mathbf{x})$ as the probability measure on*

$(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ *with density*

$$p_{\Delta_\epsilon}(\mathbf{x}' \mid \mathbf{x}) \propto \begin{cases} 0, & \mathbf{x}' \in B_\epsilon(\mathbf{x}), \\ p(\mathbf{x}'), & \textit{otherwise}, \end{cases}$$

*where* $B_\epsilon(\mathbf{x})$ *denotes the* $\epsilon$*-ball centered at* $\mathbf{x}$*. In particular,* $\mathbf{x} \notin \operatorname{supp}(\Delta_\epsilon(\mathbf{x}))$.

We are now ready to state the main definitions of PCI. Because PCI is not constrained to particular variable types, we write the following in general measure theoretic terms that support combinations of continuous and discrete variables. Below, we use the interventional law (definition 8) — a Dirac measure on deterministic causal outcomes — and marginalize over random samples of suspect sets, witness sets, and alternative values for the causal set. This results in a joint measure on the outcome under the factual and counterfactual settings.

**Definition 12 (Joint Necessity and Sufficiency Measure)** *Fix a probabilistic structural causal model* $M = \langle \mathbf{U}, \mathbf{V}, \mathbf{F}, P_{\mathbf{U}} \rangle$ *and an outcome variable* $Y$ *with domain* $\operatorname{dom}(Y)$*. Let* $p_s^\Gamma$ *and* $p_w^\Gamma$ *be the mass functions of variable selection distributions on* $2^{\mathbf{S}}$ *and* $2^{\mathbf{W}}$*, and let* $\Delta(\mathbf{s}^\star)$ *be an alternative value distribution on* $\operatorname{dom}(\mathbf{S})$*. Let* $(\mathbf{S}, \mathbf{W}) = (\mathbf{s}^\star, \mathbf{w}^\star)$ *be a factual event.*

*For a variable of interest* $X_k$*, let* $2_k^{\mathbf{S}} := \{\mathbf{C} \subseteq \mathbf{S} : X_k \in \mathbf{C}\}$*, and for each* $\mathbf{C} \subseteq \mathbf{S}$ *let* $\Delta_{\mathbf{C}}(\mathbf{s}^\star)$ *denote the pushforward of* $\Delta(\mathbf{s}^\star)$ *under the restriction map* $\mathbf{s}' \mapsto \mathbf{s}' \,|_{\mathbf{C}}$*.*[3] *For any measurable* $A \subseteq \operatorname{dom}(Y)$ *and exogenous realization* $\mathbf{u}$*, the* sufficiency *and* necessity *measures are*

$$P_k^s(A \mid \mathbf{u}) = \sum_{\mathbf{C} \in 2_k^{\mathbf{S}}} \sum_{\substack{\mathbf{T} \in 2^{\mathbf{W}} \\ \mathbf{T} \cap \mathbf{C} = \emptyset}} P(Y \in A \mid \mathbf{u}, \operatorname{do}(\mathbf{C} = \mathbf{s}^\star|_{\mathbf{C}}, \mathbf{T} = \mathbf{w}^\star|_{\mathbf{T}}))\; p_s^\Gamma(\mathbf{C})\, p_w^\Gamma(\mathbf{T}), \tag{1}$$

$$P_k^n(A \mid \mathbf{u}) = \sum_{\mathbf{C} \in 2_k^{\mathbf{S}}} \sum_{\substack{\mathbf{T} \in 2^{\mathbf{W}} \\ \mathbf{T} \cap \mathbf{C} = \emptyset}} \int P\big(Y \in A \mid \mathbf{u}, \operatorname{do}\big(\mathbf{C} = \mathbf{c}', \mathbf{T} = \mathbf{w}^\star|_{\mathbf{T}}\big)\big)\, \Delta_{\mathbf{C}}(\mathbf{s}^\star)(d\mathbf{c}')\, p_s^\Gamma(\mathbf{C})\, p_w^\Gamma(\mathbf{T}). \tag{2}$$

*The joint necessity-sufficiency measure then is given by*

$$P_k^{s,n}(A \times B) \;:=\; \int P_k^s(A \mid \mathbf{u})\, P_k^n(B \mid \mathbf{u})\, P_{\mathbf{U}}(d\mathbf{u}). \tag{3}$$

With the necessity-sufficiency measure fully defined, we are ready to characterize the causal impact of an event $X_k$. In particular, we will proceed by defining an arbitrary *causal impact function*, whose expectation over the necessity-sufficiency measure determines the causal impact. We first proceed with a formal definition of the causal impact measure, followed by several intuitive candidates for the causal impact function.

**Definition 13 (Causal Impact Function and Its Expectation)** *Let* $Y^s, Y^n \sim P_k^{s,n}$ *and let* $y^\star \in \operatorname{dom}(Y)$ *denote the factual outcome value. For any measurable causal impact function* $ci : \operatorname{dom}(Y)^3 \to \mathbb{R}$*, the* (expected) causal impact *of* $X_k$ *is defined as*

$$\mathbb{E}[\, ci(Y^s, Y^n, y^\star)] \;:=\; \iint ci(y^s, y^n, y^\star)\, P_k^{s,n}(dy^s, dy^n). \tag{4}$$

3. For an assignment $\mathbf{x}$ to a collection of variables $\mathbf{X}$ and any subset $\mathbf{Y} \subseteq \mathbf{X}$, the restriction $\mathbf{x}\,|_{\mathbf{Y}}$ denotes the sub-assignment of $\mathbf{x}$ to the variables in $\mathbf{Y}$.

**Example 6 (Probability of Necessary and Sufficient Causation for Binary Outcomes)** *Assume the outcome $Y$ is binary with $\mathrm{dom}(Y) = \{0, 1\}$, and let $y^\star = 1$ denote the factual outcome. Let $Y^s$ and $Y^n$ denote the sufficiency and necessity outcome variables. Define the causal impact score*

$$ci(y^s, y^n, y^\star) = \mathbb{I}\{y^n \neq y^\star\}\, \mathbb{I}\{y^s = y^\star\}.$$

*Then the expected causal impact reduces to*

$$\mathbb{E}[ci(Y^s, Y^n, 1)] = \mathbb{P}(Y^n = 0,\, Y^s = 1),$$

*which conceptually coincides with Pearl's probability of necessity and sufficiency (PNS). The alignment is* exact *when $\mathbf{S} = \{X_k\}$, $\mathbf{W} = \emptyset$, and $\mathrm{dom}(X_k) = \{0, 1\}$ where, factually, $x_k^\star = 1$.*

**Example 7 (Absolute Difference Impact Score)** *We can develop an analogous measure for continuous outcomes where we increase attribution with distance from the factual value $y^\star$ in the necessity world and decrease impact with distance to the factual value in the sufficiency world.*

$$ci(y^s, y^n, y^\star) = |y^n - y^\star| - |y^s - y^\star|.$$

## 4. Examples and comparison to SHAP and Causal SHAP

Shapley values have become one of the most widely used tools for explaining the predictions of complex machine learning models. Grounded in cooperative game theory, they provide a principled decomposition of a model's prediction into additive contributions from individual features. However, when the goal is *causal responsibility attribution* — explaining not just what a model predicts but why an outcome occurred — standard Shapley values suffer from fundamental limitations that are not always made explicit in the literature.

update section description when done

In this section we inspect these limitations carefully: we work throughout with a simple but instructive running example, simple enough to permit exact analytic computation while being rich enough to expose the core conceptual issues. We then examine how Causal Shapley values (**?**) partially address these issues, where they succeed, and where fundamental gaps remain.

**Example 8 (Signal with mediation)** *We work with the following **generative model**, where $X$ sends a signal, $M$ mediates it with some noise, and $Y$ is the final outcome, which is also noisy:*

$$X \sim \mathcal{N}(0.5,\ 0.25) \tag{5}$$

$$M = X + \varepsilon_M, \quad \varepsilon_M \sim \mathcal{N}(0,\ 0.1) \tag{6}$$

$$Y = M + \varepsilon_Y, \quad \varepsilon_Y \sim \mathcal{N}(0,\ 0.1) \tag{7}$$

*The causal graph is $X \to M \to Y$, a simple mediation chain. All noise terms are independent of each other and of $X$. Since everything is jointly Gaussian, all conditional expectations are exact linear functions — no approximations are required.*

*Let $R(X \rightsquigarrow Y)$ denote the causal responsibility of $X$ for $Y$. The **qualitative causal responsibility desiderata** for this example are rather simple:*

$$\begin{aligned}
&\textbf{\textit{DXY}} \quad R(X \rightsquigarrow Y) > 0 \\
&\textbf{\textit{DMY}} \quad R(M \rightsquigarrow Y) > 0 \\
&\textbf{\textit{DXM}} \quad R(X \rightsquigarrow M) > 0
\end{aligned}$$

$$\begin{aligned}
&\textbf{\textit{DYX}} \quad R(Y \rightsquigarrow X) = 0 \\
&\textbf{\textit{DYM}} \quad R(Y \rightsquigarrow M) = 0 \\
&\textbf{\textit{DMX}} \quad R(M \rightsquigarrow X) = 0
\end{aligned}$$

*Two somewhat more subtle desiderata are as follows:*

$$\textbf{\textit{DMXY}} \quad R(M \rightsquigarrow Y) > R(X \rightsquigarrow Y)$$

$$\textbf{\textit{DMYX}} \quad R(M \rightsquigarrow Y) > R(X \rightsquigarrow Y)$$

*they capture the intuition that responsibility should be higher for variables that are closer to the outcome in the causal graph.*
*The key **distributional facts** that we will use are as follows:*

$$\mathbb{E}[X] = \mathbb{E}[M] = \mathbb{E}[Y] = 0.5, \tag{8}$$
$$\big(\mathit{Var}(X),\, \mathit{Var}(M),\, \mathit{Var}(Y)\big) = (0.25,\, 0.35,\, 0.45), \tag{9}$$
$$\big(\mathit{Cov}(X,M),\, \mathit{Cov}(X,Y),\, \mathit{Cov}(M,Y)\big) = (0.25,\, 0.25,\, 0.35). \tag{10}$$

*For each choice of target variable, the **optimal predictor** (minimizing squared error) is the conditional expectation. These can be computed exactly for all three choices of target variable.*

$$f_Y(X,M) = \mathbb{E}[Y \mid X, M] = M \qquad (\{X,M\} \Rightarrow Y)$$
$$f_M(X,Y) = \mathbb{E}[M \mid X, Y] = 0.5X + 0.5Y \qquad (\{X,Y\} \Rightarrow M)$$
$$f_X(M,Y) = \mathbb{E}[X \mid M, Y] = 0.5 + \frac{5}{7}(M - 0.5) \qquad (\{M,Y\} \Rightarrow X)$$

*add fn with derivations?*

Before we compute SHAP values for the case at hand, let's talk through the definition of **Shapley values**.

**Definition 14 (SHAP)** *A cooperative game is a pair $(N, v)$, where $N = \{1, \ldots, n\}$ is the set of players and $v : 2^N \to \mathbb{R}$ is the* characteristic function *mapping each subset (coalition) $S \subseteq N$ to the value that coalition can produce. The Shapley value (**?**) is the unique attribution satisfying four axioms:*

1. ***Efficiency***: $\sum_i \phi_i = v(N) - v(\emptyset)$

2. ***Symmetry***: *If* $v(S \cup \{i\}) = v(S \cup \{j\})$ *for all* $S$*, then* $\phi_i = \phi_j$

3. ***Dummy***: *If* $v(S \cup \{i\}) = v(S)$ *for all* $S$*, then* $\phi_i = 0$

4. ***Linearity***: *For two games* $v_1, v_2$*:* $\phi_i(\alpha_1 v_1 + \alpha_2 v_2) = \alpha_1 \phi_i(v_1) + \alpha_2 \phi_i(v_2)$

*The unique solution is:*

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!\,(|N| - |S| - 1)!}{|N|!} \big[v(S \cup \{i\}) - v(S)\big] \tag{11}$$

*The weight* $\frac{|S|!(|N|-|S|-1)!}{|N|!}$ *equals the probability that, in a uniformly random ordering of all players, exactly the members of* $S$ *precede* $i$*. Equivalently,* $\phi_i$ *is the average marginal contribution of* $i$ *across all orderings. Lundberg and Lee (2017) apply Shapley values to explain model predictions by taking players to be input features, and the characteristic function to be the expected model output given a subset of features:*

$$v(S) = \mathbb{E}_{X_{\bar{S}}}\big[f(X_S = x_S,\, X_{\bar{S}})\big] \tag{12}$$

*where* $\bar{S} = N \setminus S$ *and missing features are marginalized.*

Get back to this and illustrate:
One crucial observation here is that SHAP explains predictions, not outcomes. If $f(x) = \mathbb{E}[Y \mid X = x]$ (as in regression), then SHAP explains $\mathbb{E}[Y \mid X = x] - \mathbb{E}[Y]$. The residual $Y - f(x)$ — the noise — is entirely invisible to SHAP by construction.

Now consider a situation in which $X = 1, M = 1, Y = 1$. For each of those outcomes we will use our straighforward causal resposbility attribution desiderata and see how they align with SHAP values. First, let's compute the SHAP values for $X$ and $M$ with respect to the prediction of $Y$.

### 4.1. Target $Y$, Features $\{X, M\}$

$$v(\emptyset) = \mathbb{E}[M] = 0.5 \tag{13}$$
$$v(\{X\}) = \mathbb{E}[M \mid X = 1] = 1 \tag{14}$$
$$v(\{M\}) = \mathbb{E}[M \mid M = 1] = 1 \tag{15}$$
$$v(\{X, M\}) = f_Y(1, 1) = 1 \tag{16}$$

Applying equation (11) with $|N| = 2$:

$$\phi_X = \tfrac{1}{2}(1 - 0.5) + \tfrac{1}{2}(1 - 1) = 0.25 \qquad \phi_M = \tfrac{1}{2}(1 - 0.5) + \tfrac{1}{2}(1 - 1) = 0.25 \tag{17}$$

So

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